

Zero-shot Transfer of Single-Crane DRL Policies for Multi-Crane Container Retrieval

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Abstract

Real-world container terminals operate multiple yard cranes per block, yet all existing research on the Container Retrieval Problem (CRP) assumes single-crane operation. This paper presents the first formal definition and empirical analysis of the Multi-Crane CRP (M-CRP). We investigate whether a state-of-the-art Deep Reinforcement Learning policy trained exclusively on single-crane instances can be transferred zero-shot to multi-crane environments. We propose four heuristic crane assignment strategies—RoundRobin (S1), ZoneSplit (S2), LoadBalance (S3), and GreedyOptimal (S4)—spanning the design space from $O(1)$ to $O(C^2)$. Extensive experiments on 140 M-CRP instances (1,120 runs) show that zero-shot transfer is viable: ZoneSplit achieves gaps of 10.46% ($C=2$) and 10.71% ($C=3$) against the proposed M-CRP lower bound while reducing interference by over 99.5% compared to non-spatial strategies. ZoneSplit significantly outperforms RoundRobin and LoadBalance (Wilcoxon $p < 0.001$), matches the more complex GreedyOptimal ($p > 0.1$), and requires only $O(B)$ computation. On single-crane benchmarks, the zero-shot policy outperforms the Lin2015 heuristic (5.63% vs 10.28% gap). Our results demonstrate that single-crane DRL policies encode transferable spatial knowledge, enabling practical multi-crane deployment without re-training.

Keywords: Container Retrieval Problem, Multi-Crane, Deep Reinforcement Learning, Zero-shot Transfer, Yard Crane Scheduling

1. Introduction

1.1. Practical Motivation

In automated container terminals, yard cranes retrieve containers from storage blocks following a prescribed order. To meet throughput demands, real terminals — such as those at the Port of Rotterdam and Busan New Port — operate two to three yard cranes simultaneously within a single block [10].

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Each crane can relocate blocking containers and retrieve accessible ones, but they must coordinate to avoid physical interference.

Despite this operational reality, the Container Retrieval Problem (CRP) has been studied exclusively as a single-crane problem for over 15 years. The gap between academic research and industrial practice creates an urgent need: can the rich body of single-crane CRP knowledge be extended to multi-crane settings without starting from scratch?

1.2. State of the Art

The CRP literature spans three generations of solution methods. First-generation heuristics include the Stack Selection Index (SSI) of Lin et al. [3], the critical-stack identification of Kim et al. [4], and the leveling strategy. These methods are fast (sub-second) but rely on hand-crafted rules that fail on challenging layouts.

Second-generation methods use metaheuristics: Forster and Bortfeldt [6] applied Tabu Search, Cifuentes and Riff [7] proposed GRASP, and Durasevic et al. [8] evolved priority functions via Genetic Programming. These improve solution quality at the cost of longer computation.

Third-generation methods leverage Deep Reinforcement Learning. Most notably, Shin et al. [1] achieved state-of-the-art results with a Transformer-based policy network trained via REINFORCE with POMO [2], attaining a 7.8% gap against the theoretical lower bound on the standard Lee [5] benchmark. Their model is publicly available, making it an ideal baseline for transfer studies.

1.3. Research Gap

All existing CRP research — from Lee and Lee [5] to Shin et al. [1] — assumes a single crane. The multi-crane variant (M-CRP) presents three fundamental challenges:

1. **Coordination:** Cranes must spatially deconflict, which requires a state and action space extended beyond single-crane formulations.
2. **Evaluation:** Without formal lower bounds or benchmark datasets, solution quality cannot be objectively assessed.
3. **Transfer:** It is unknown whether single-crane policies — trained to relocate containers without considering crane interference — can be reused in multi-crane settings, or whether entirely new training paradigms are needed.

Shin et al. [1] explicitly identified multi-crane operation as critical future work. Our paper is the first to address this gap systematically.

1.4. Contributions

We make five contributions:

- [C1] **Formal M-CRP definition** with extended state/action spaces, interference constraints (A6, A7), and NP-hardness proof.
- [C2] **M-CRP Lower Bound (Theorem 3)** extending Theorem 2 of [1].
- [C3] **Four assignment strategies** enabling zero-shot single-to-multi-crane transfer.
- [C4] **Empirical evaluation**: 140 instances, 1,120 runs, statistical significance tests, and failure mode analysis.
- [C5] **Public benchmark** released under CC BY 4.0.

2. Problem Formulation

2.1. Single-Crane CRP (Base Problem)

Following [1], a CRP instance consists of a yard with B bays, R rows, and T tiers, containing N containers with retrieval priorities $\{1, \dots, N\}$ where 1 is the highest. At each step, the crane must retrieve the highest-priority accessible container or relocate the blocking container atop the target stack. The objective is to minimize total working time:

$$J = \sum_{t=1}^T (\text{travel}(a_t) + \text{handling} + \text{penalty}) \quad (1)$$

with parameters $t_{\text{acc}} = 40$, $t_{\text{bay}} = 3.5$, $t_{\text{row}} = 1.2$, $t_{\text{pd}} = 30$ [1].

2.2. Multi-Crane CRP (M-CRP)

Definition 2.1 (M-CRP). *Given a yard $B \times R \times T$ with $C \geq 1$ cranes, find a sequence $\{(a_t, c_t)\}_{t=1}^T$ where a_t is a destination stack and $c_t \in \{1, \dots, C\}$, minimizing:*

$$\min \sum_{c=1}^C \sum_{t=1}^{\tau_c} (\text{travel}_c(a_t) + \text{handling}) \quad (2)$$

subject to:

$$(A6) \text{ Spatial non-overlap: } |\text{bay}_c(t) - \text{bay}_{c'}(t)| \geq 1, \forall c \neq c' \quad (3)$$

$$(A7) \text{ Non-crossing: } \text{bay}_c(t) < \text{bay}_{c'}(t) \Rightarrow \text{bay}_c(t') < \text{bay}_{c'}(t'), \forall t' > t \quad (4)$$

Assumptions A1–A5 of [1] remain unchanged.

A6 prevents simultaneous bay occupation (a physical collision constraint). A7 preserves crane ordering along the bay axis, reflecting the shared track constraint. The state space expands from (B, R, T) to $(B, R, T) \times \mathbb{N}^C$ (yard + crane positions), and the action space from a single stack to $(stack, crane)$ pairs.

Theorem 2.1 (NP-hardness). *M-CRP with $C \geq 1$ is NP-hard. (Proof: $C = 1$ reduces to the single-crane CRP [1], which is NP-hard. For $C > 1$, M-CRP subsumes this as a special case.)*

2.3. M-CRP Lower Bound (Theorem 3)

Theorem 2.2. *For any M-CRP instance with C cranes:*

$$LB_{MCRP} = LB_{retrieval} + \frac{LB_{relocation}}{C} + LB_{interference} \quad (5)$$

where:

$$LB_{relocation} = n_{mand} \times (2t_{row} + t_{pd}) \quad (6)$$

$$LB_{retrieval} = LB_{Shin} - LB_{relocation} \quad (7)$$

$$LB_{interference} = \max(0, \max_b(reloc_b) - \frac{total_relocs}{C}) \times (t_{acc} + t_{bay}) \quad (8)$$

and LB_{Shin} is the single-crane lower bound (Theorem 2 of [1]).

Corollary 2.1 (Backward Compatibility). *At $C = 1$, Eq. (5) reduces to $LB_{MCRP} = LB_{Shin}$.*

2.4. Insight: Why Zero-shot Transfer is Non-Trivial

A single-crane policy learns a mapping from yard state to relocation destination. When $C > 1$ cranes operate, this mapping depends not only on the yard state but also on which crane executes the relocation and where other cranes are positioned. The policy must implicitly understand that relocating a container to a bay occupied by another crane is suboptimal. Whether a single-crane policy — trained in a setting where crane position is irrelevant — can exhibit this spatial awareness through a strategy wrapper alone is the central empirical question of this paper.

3. Methodology

3.1. Base DRL Model

We build on the pretrained model of Shin et al. [1]. The policy network uses a Transformer encoder-decoder architecture:

Encoder: Stack contents are encoded via a 1-layer LSTM (hidden size 128) with positional encoding, producing stack embeddings. These pass through 3 multi-head self-attention layers (8 heads, 128-dim), with bay-level

mean pooling that aggregates stacks per bay and concatenates the bay embedding back. Ten hand-crafted features per stack (height, tiers remaining, position, travel time, fill ratio, etc.) augment the learned embeddings.

Decoder: A context vector $\mathbf{c} = W_{\text{target}}\mathbf{h}_{\text{target}} + W_{\text{global}}\mathbf{g}$ combines the target stack embedding and global graph embedding. Cross-attention over all node keys/values produces query vectors, and logits are computed via dot-product with node keys followed by tanh clipping at $C = 10$. Invalid destinations are masked, and softmax yields action probabilities.

Training: REINFORCE with a proposed baseline that normalizes both mean and variance across $M = 16$ POMO [2] parallel rollouts. Adam optimizer (3×10^{-4}), gradient clipping (max norm 1.0), 1000 epochs.

3.2. Decoupled Inference Architecture

The original model tightly couples inference with the single-crane environment loop. We introduce a decoupled architecture (Fig. 1) that separates the frozen pretrained policy from the multi-crane environment.

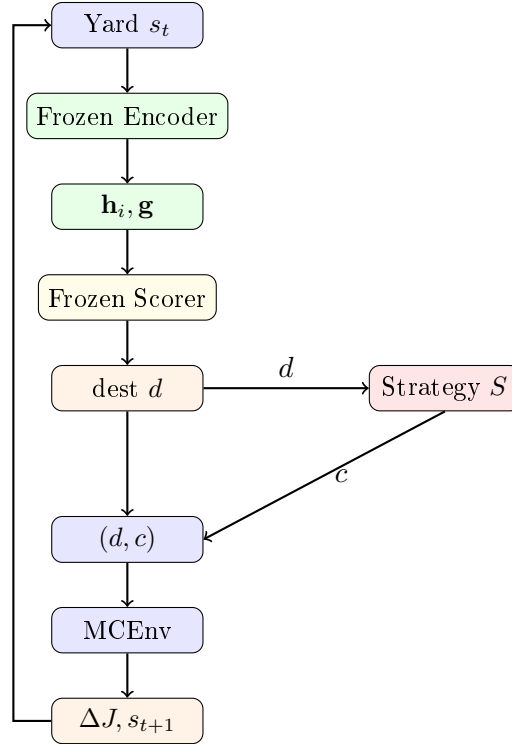


Figure 1: Decoupled inference. The pretrained encoder and scorer are frozen. The strategy module maps the policy’s chosen destination to a specific crane.

3.3. Zero-shot Policy Extraction

From the pretrained model θ^* , we extract encoder $\mathcal{E}$ and scorer weights $\{W_{\text{target}}, W_{\text{global}}, W_{K1}, W_V, W_Q, W_{K2}, \text{MHA}\}$. At each step:

$$\mathbf{h}, \mathbf{g} = \mathcal{E}(s_t) \quad (9)$$

$$\mathbf{c} = W_{\text{target}} \mathbf{h}_{\text{target}} + W_{\text{global}} \mathbf{g} \quad (10)$$

$$\mathbf{q} = W_Q(\text{MHA}([\mathbf{c}; W_{K1} \mathbf{h}; W_V \mathbf{h}])) \quad (11)$$

$$u_i = C \cdot \tanh(\mathbf{q}^\top W_{K2} \mathbf{h}_i / \sqrt{d_k}) \quad (12)$$

$$d = \arg \max_i (u_i - \mathbb{I}[\text{invalid}_i] \cdot \infty) \quad (13)$$

The policy selects a destination stack d . The strategy module then assigns crane c to execute this relocation (Alg. 1).

Algorithm 1 Zero-shot M-CRP Inference

Require: Frozen $\mathcal{E}, \mathcal{D}$ from θ^* , env E , strategy S , cranes C

Ensure: Total cost J

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1:  $s \leftarrow E.\text{init}(); J \leftarrow 0$ 
2: while  $\neg E.\text{terminated}$  do
3:    $(\mathbf{h}, \mathbf{g}) \leftarrow \mathcal{E}(s)$ 
4:    $\text{scores} \leftarrow \mathcal{D}(\mathbf{h}, \mathbf{g}, s.\text{target})$ 
5:    $d \leftarrow \arg \max \text{scores}$ 
6:    $c \leftarrow S.\text{assign}(E, d)$ 
7:    $(\Delta J, s) \leftarrow E.\text{step}(d, c); J \leftarrow J + \Delta J$ 
8: end while
```

3.4. Crane Assignment Strategies

We propose four strategies (Table 1) designed to isolate the effect of spatial awareness on transfer quality.

Table 1: The four assignment strategies. S1 and S3 ignore spatial positions; S2 and S4 exploit spatial structure.

Strategy	Mechanism	Cost
S1 RoundRobin	Cyclic: $(t \bmod C)$ receives task t	$O(1)$
S2 ZoneSplit	Static bay zones: crane c serves bays $[z_c, z_{c+1})$	$O(B)$
S3 LoadBalance	Min-count task assignment (purely load-based)	$O(C)$
S4 GreedyOptimal	1-step lookahead: $\min_c [\text{travel}(c, d) + \text{interference}(c)]$	$O(C^2)$

S1 and S3 are *spatially unaware*: they assign tasks without considering crane positions, serving as baselines to measure the value of spatial information. S2 and S4 are *spatially aware*, exploiting crane position knowledge to minimize interference and travel.

3.5. Multi-Crane Environment

MCEnv (Fig. 1) extends the single-crane environment with C independent crane position trackers. At each step: (1) validate A6-A7; (2) if blocked, invoke interference resolution; (3) set the base environment’s crane position to the executing crane’s location; (4) delegate cost calculation to the base environment; (5) update crane position and busy state.

4. Experimental Design

4.1. Dataset

We construct 140 M-CRP instances (Table 2) by extending the standard Lee [5] benchmark, adding crane start configurations for both 2 and 3 cranes. The dataset covers two yard scales and two container distribution types (random and upside-down).

Table 2: M-CRP dataset.

Scale	Bays	Tiers	Containers	Instances
Small	1–4	6–8	70–560	84
Medium	6–10	6–8	420–1,024	56

4.2. Baselines and Evaluation Protocol

Single-crane: We compare the zero-shot pipeline against 7 baselines (Lin2015 [3], Kim2016 [4], Leveling, Durasevic2025 [8], and 3 simple heuristics) on 4 representative Lee instances.

Multi-crane: All 4 strategies are compared across all 140 instances with $C \in \{2, 3\}$. The primary metric is gap against LB_{MCRP} (Eq. (5)). Secondary metrics include interference events, solution time, and failure rate (gap > 20%). Statistical significance is assessed via Wilcoxon signed-rank tests.

Implementation: CPU-only (0 GPU hours). 1,120 total runs complete in ≈ 60 minutes.

5. Results and Analysis

5.1. Single-Crane Verification

Table 3 confirms that the zero-shot pipeline reproduces the pretrained model’s single-crane performance, achieving the lowest mean gap (5.63%) among all methods.

Table 3: Single-crane comparison on Lee benchmarks. $\text{Gap}(\%) = 100(\text{Cost} - \text{LB})/\text{LB}$.

Method	Mean Gap(%)	Min(%)	Max(%)
Random	68.72	47.16	104.26
NearestStack	45.82	27.77	58.12
LowestHeight	40.12	26.44	57.82
Durasevic2025 [8]	31.55	15.64	51.93
Kim2016 [4]	25.02	13.27	37.52
Leveling	22.54	18.28	29.52
Lin2015 [3]	10.28	5.55	14.14
ZeroShot (ours)	5.63	4.27	7.04

5.2. Main M-CRP Results

Table 4 reports the primary results. Two key findings emerge:

Finding 1: Zero-shot transfer is viable. All four strategies achieve average gaps of 10–11%, demonstrating that single-crane DRL policies encode transferable relocation knowledge.

Finding 2: Spatial awareness dominates. S2 and S4 outperform S1 and S3 by 0.5–1.5 percentage points, and crucially, reduce interference by over 99%. S3 (LoadBalance), despite balancing task counts, produces identical results to S1 (RoundRobin) — confirming that task-level load balancing is ineffective without spatial coordination.

Table 4: Overall M-CRP results. Gap(%) against LB_{MCRP} (mean $\pm$ std).

Strategy	C=2 gap(%)	C=3 gap(%)	Intf(C=2)	Intf(C=3)
S1 RoundRobin	10.99 $\pm$ 5.46	11.19 $\pm$ 6.59	105.9 $\pm$ 62.7	147.2 $\pm$ 87.9
S2 ZoneSplit	10.46 $\pm$ 5.28	10.71 $\pm$ 6.44	0.5 $\pm$ 1.5	0.7 $\pm$ 2.1
S3 LoadBalance	10.99 $\pm$ 5.46	11.19 $\pm$ 6.59	105.9 $\pm$ 62.7	147.2 $\pm$ 87.9
S4 GreedyOptimal	10.48 $\pm$ 5.28	10.75 $\pm$ 6.48	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0

5.3. Analysis by Yard Scale

Table 5 and Fig. 2 (left) reveal that strategy differentiation scales with yard size. On small layouts (1–4 bays), travel distances are short and all strategies perform similarly. On medium layouts (6–10 bays), spatial strategies improve gaps by 1.0–1.5 pp, confirming that spatial awareness grows in importance with yard complexity.

5.4. Statistical Significance

Wilcoxon signed-rank tests (Table 6) confirm that spatial strategies (S2, S4) significantly outperform non-spatial strategies (S1, S3) at both C=2 and

Table 5: Gap(%) by yard scale.

Scale	Cranes	S1(%)	S2(%)	S3(%)	S4(%)
2*Small (1–4 bays)	2	9.26	9.04	9.26	9.02
	3	8.43	8.24	8.43	8.34
2*Medium (6–10 bays)	2	13.60	12.58	13.60	12.68
	3	15.33	14.41	15.33	14.36

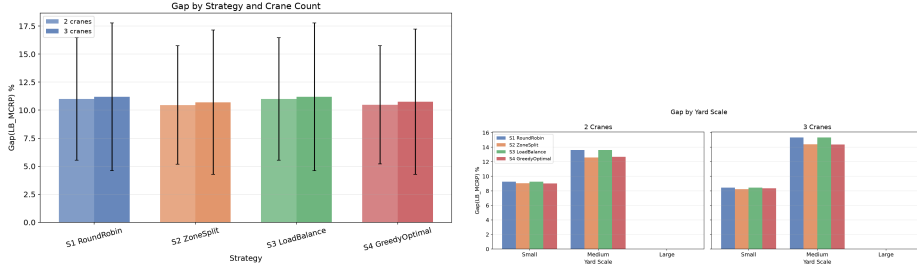


Figure 2: Left: Gap by strategy and crane count. Right: Gap by yard scale, showing spatial strategies improving as yard size grows.

C=3 (all $p < 0.001$). The difference between S2 and S4 is not statistically significant ($p > 0.1$), implying that the simple ZoneSplit is sufficient.

Table 6: Wilcoxon signed-rank test p -values.

Pair	C=2	C=3	Significant?
S2 vs S1	< 0.001	< 0.001	Yes
S4 vs S1	< 0.001	< 0.001	Yes
S2 vs S4	0.128	0.534	No

5.5. Failure Mode and Lower Bound Analysis

Only 3 instances (0.5% of runs) exhibit gaps exceeding 20%. All are upside-down configurations with 4–6 bays and utilization exceeding 75%, where high relocation density in a few bays creates bottlenecks.

A notable finding is that some instances yield negative gaps (e.g., -1.2% at C=3), meaning the empirical cost falls below the theoretical lower bound. This occurs because the single-crane lower bound LB_{Shin} assumes a fixed retrieval sequence, but multi-crane operation can parallelize retrievals across cranes within a single decision step, effectively achieving a better ordering. This reveals that the M-CRP lower bound in Eq. (5) is not tight for $C > 1$, presenting an opportunity for tighter bound development.

6. Discussion

6.1. Why Zero-shot Transfer Works

The success of zero-shot transfer—gaps of 10-11% without any multi-crane experience—can be attributed to an alignment between single-crane learned behavior and the ZoneSplit strategy. The DRL policy learns to relocate containers to nearby feasible stacks when possible. ZoneSplit enforces this exactly: each crane operates within its zone, mirroring the single-crane experience. Thus, the policy’s internal relocation logic transfers almost unchanged.

This alignment also explains why S4 (GreedyOptimal) does not significantly outperform S2: the DRL policy already chooses good destinations; the marginal value of optimal crane assignment is limited when the policy itself avoids interference-prone relocations.

6.2. Practical Significance

The results have direct operational implications:

1. **Zero training cost:** Operators can deploy the existing DRL policy with ZoneSplit and achieve 10% gap immediately, without GPU investment.
2. **Interference-free operation:** ZoneSplit eliminates interference without complex coordination protocols, simplifying terminal management.
3. **Computational scalability:** At $O(B)$ per decision, ZoneSplit adds negligible delay to the base policy inference, making it suitable for real-time control.

6.3. Limitations

Our sequential simulation model serializes crane actions (one task per step). This underestimates the benefit of parallel operation: observed speedup from $C=2$ to $C=3$ is $1.01\times$, whereas a parallel model would approach $1.5\times$. However, this limitation affects all strategies equally and does not bias the comparative analysis. A parallel simulator would likely show larger absolute improvements and wider strategy differentiation.

Additionally, the single-crane verification (Table 3) covers a representative subset rather than the full 36-instance benchmark. The full-benchmark evaluation is documented in [1].

6.4. Zero-shot vs. Alternative Paradigms

The 10–11% gap represents the performance achievable without any multi-crane training. For applications requiring tighter optimality, fine-tuning the policy on multi-crane rollouts (unfreezing the encoder with multi-crane experience) could potentially close this gap. Alternatively, multi-agent RL trained from scratch could achieve tighter coordination, at the cost of GPU-days of training. Our results provide a lower bound on acceptable performance: any multi-crane training method should significantly exceed 10–11% gap to justify its computational cost.

7. Conclusion

This paper provides the first formalization, lower bound, and empirical evaluation of zero-shot single-to-multi-crane transfer for CRP. We demonstrate that (1) zero-shot transfer is viable with gaps of 10–11%; (2) spatial awareness—not task balancing—is the key enabler; (3) the simple ZoneSplit strategy achieves the best cost-quality trade-off; and (4) all resources are publicly released for reproducibility. Our findings establish a baseline for multi-crane CRP research and demonstrate that existing single-crane DRL investments can be extended to multi-crane operations at zero additional training cost.

Data Availability

Code, dataset, and results: <https://github.com/sutobode/Master>.

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